

A compact quotient-coordinate reduction for the original Kalton operator-space pair of arXiv:2106.15022

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Status. Partial result, likely valid; the exact source question remains open after the reduction below.

1 Source question and setup

Braga's Theorem 1.8 constructs operator spaces X and Y such that X embeds isomorphically but not completely isomorphically into Y , while their bounded subsets are almost completely coarsely equivalent [1]. The paragraph following the theorem asks whether the specific examples also satisfy that X does not *almost* completely isomorphically embed into Y .

Theorem 1.8. *There are operator spaces X and Y so that*

- (1) *X linearly isomorphically embeds into Y ,*
- (2) *X does not completely isomorphically embed into Y , and*
- (3) *the bounded subsets of X and Y are almost completely coarsely equivalent.*

We point out that we do not know if the examples of X and Y provided by Theorem 1.8 are such that X does not *almost* completely isomorphically embed into Y . Also, the examples in Theorem 1.8 are nonseparable.

Figure 1: Theorem 1.8 and the source question on PDF page 4 of [1].

For the explicit construction, put $H = R \cap C$ and let $Q : L_\infty[0, 1] \rightarrow H$ be the complete quotient dual to the standard complete embedding $R + C \rightarrow L_1[0, 1]$. On the vector space $L_\infty[0, 1]$, define

$$\|y\|_{Y_j} = \max\{2^{-j}\|y\|_\infty, \|Qy\|_H\}, \quad \mathcal{Z}(Q) = \left(\bigoplus_{j \geq 1} Y_j \right)_{\ell_1}.$$

The source pair is $Y = \mathcal{Z}(Q)$ and $X = H \oplus \ker \tilde{Q}$. Thus it is enough to obstruct an almost complete embedding $H \rightarrow \mathcal{Z}(Q)$.

Let

$$W = \left(\bigoplus_{j \geq 1} \text{MIN}(L_\infty[0, 1]) \right)_{\ell_1}, \quad U = \left(\bigoplus_{j \geq 1} H \right)_{\ell_1}.$$

The canonical coordinate maps are

$$P_1((y_j)_j) = (2^{-j}y_j)_j \in W, \quad P_2((y_j)_j) = (Qy_j)_j \in U.$$

The map (P_1, P_2) is a complete isomorphic embedding of $\mathcal{Z}(Q)$ into $W \oplus_\infty U$ with distortion at most 2. This follows directly from the coordinate maximum defining Y_j and the universal construction of the operator-space ℓ_1 sum.

2 Proof intuition

The quotient part of any linear map $H \rightarrow \mathcal{Z}(Q)$ is compact. Indeed, a noncompact coordinate $H \rightarrow H$ is bounded below on an infinite-dimensional Hilbert subspace. Its lift to L_∞ would then yield, using the quotient Q and an orthogonal projection in H , a bounded projection of L_∞ onto a copy of ℓ_2 , which is impossible.

Compactness lets us make all quotient coordinates simultaneously as small as needed on an infinite-dimensional subspace. At matrix level n we make the scalar norm $O(1/n)$; the standard estimate $\|S_n\| \leq n\|S\|$ then makes the n th amplification negligible. Hence the minimal L_∞ coordinates alone must preserve the lower matrix estimate.

3 Compact quotient-coordinate theorem

Lemma 1. *Let $T : H \rightarrow L_\infty[0, 1]$ be bounded. Then $QT : H \rightarrow H$ is compact.*

Proof. Suppose $A = QT$ is not compact. Since both its domain and range are Hilbert spaces, there is an infinite-dimensional closed subspace $E \subset H$ and $c > 0$ such that $\|Ax\| \geq c\|x\|$ on E . Thus $A(E)$ is closed. Let $P_{A(E)}$ be the orthogonal projection of H onto $A(E)$. Then

$$\Pi = T|_E (A|_E)^{-1} P_{A(E)} Q : L_\infty[0, 1] \longrightarrow T(E)$$

is bounded and satisfies $\Pi(Tx) = Tx$ for $x \in E$. Hence $T(E)$ is a complemented copy of ℓ_2 in $L_\infty[0, 1]$, contradicting the standard complemented-subspace theorem for L_∞ spaces used in the proof of Proposition 4.6 of [1]. Therefore A is compact. $\square$

Theorem 1. *Let $u : H \rightarrow \mathcal{Z}(Q)$ be bounded and write u_j for its j th coordinate. Then*

$$V_u : H \longrightarrow \ell_1(H), \quad V_u x = (Qu_j x)_{j \geq 1},$$

is compact.

Proof. The definition of the $\mathcal{Z}(Q)$ norm gives

$$\sum_j \|Qu_j x\| \leq \|ux\|_{\mathcal{Z}(Q)},$$

so V_u is bounded. By Lemma 1, every coordinate $Qu_j : H \rightarrow H$ is compact.

Since H is reflexive, V_u is weakly compact. The standard weak compactness criterion in the vector-valued sum $\ell_1(H)$ says that a relatively weakly compact set has uniformly small coordinate tails (the coordinate sets are weakly compact because H is reflexive). Consequently, if p_N is the projection onto the first N coordinates, then $\|(I - p_N)V_u\| \rightarrow 0$. Each $p_N V_u$ is compact, being a finite tuple of compact operators. Hence V_u is a norm limit of compact operators and is compact. $\square$

4 Reduction of the source question

Theorem 2. *If $H = R \cap C$ almost completely isomorphically embeds into $\mathcal{Z}(Q)$, then H almost completely isomorphically embeds into $W = (\bigoplus_j \text{MIN}(L_\infty[0, 1]))_{\ell_1}$.*

Proof. Assume there are linear maps $u^{(n)} : H \rightarrow \mathcal{Z}(Q)$ and $D \geq 1$ such that

$$D^{-1}\|a\|_{M_n(H)} \leq \|u_n^{(n)}(a)\|_{M_n(\mathcal{Z}(Q))} \leq D\|a\|_{M_n(H)}$$

for every n and $a \in M_n(H)$.

Fix n . The operator $S_n = P_2 u^{(n)} : H \rightarrow U$ is compact by Theorem 1. Approximate it in norm by a finite-rank operator and take an infinite-dimensional subspace $F_n \subset H$ in the kernel of that approximant. We may arrange

$$\|S_n|_{F_n}\| \leq (8Dn)^{-1}.$$

Because $H = R \cap C$ is unitarily homogeneous, there is a complete isometry $J_n : H \rightarrow F_n$. Set $T_n = P_1 u^{(n)} J_n : H \rightarrow W$.

For any bounded linear map between operator spaces, $\|S_n^{(n)}\| \leq n\|S_n\|$. Hence the n th amplification of $P_2 u^{(n)} J_n$ has norm at most $(8D)^{-1}$. On the other hand, the complete distortion-2 coordinate embedding (P_1, P_2) and the lower estimate for $u_n^{(n)}$ force

$$\|(T_n)_n(a)\| \geq (2D)^{-1}\|a\|$$

after harmlessly weakening constants. The upper estimate is $\|(T_n)_n(a)\| \leq D\|a\|$. Thus $((T_n)_n)_n$ is an equi-isomorphic family, which is precisely an almost complete isomorphic embedding of H into W . $\square$

5 Verification and limitations

The compactness argument is independent of the later construction in [2]. That paper proves the desired existential strengthening using the different quotient $\text{MAX}(L_1) \rightarrow \text{MIN}(\ell_2)$; it does not establish nonembedding for the exact $L_\infty \rightarrow R \cap C$ pair above.

Theorem 2 is only a necessary reduction. This packet does not prove that $R \cap C$ fails to almost completely embed into W . That uniform minimal-sum obstruction is the remaining problem. The main point for human verification is the complete distortion bound for (P_1, P_2) under the standard operator-space ℓ_1 -sum convention.

Local lightweight indexes, the source of arXiv:2211.11854, and bounded web searches for the exact phrase and close variants were checked on 19 July 2026. No explicit resolution of the exact original pair or this reduction was found.

References

- [1] B. M. Braga, *Towards a theory of coarse geometry of operator spaces*, arXiv:2106.15022; Israel J. Math. 259 (2024), 527–558.
- [2] B. M. Braga and T. Oikhberg, *Coarse geometry of operator spaces and complete isomorphic embeddings into ℓ_1 and c_0 -sums of operator spaces*, arXiv:2211.11854; Math. Z. 304 (2023), Article 52.
- [3] F. Albiac and N. J. Kalton, *Topics in Banach Space Theory*, 2nd ed., Springer, 2016.